

Secure-Tunnel: Post-Quantum Cryptographic Overlay for UAV C2 Links

Results and Analysis Section — v2.0 (2026-03-03)

Draft compilation for review

TABLE I
ALGORITHM ALIASES AND NIST SECURITY LEVELS

Alias	Full Name	Level	Type
ML512	ML-KEM-512	L1	KEM
ML768	ML-KEM-768	L3	KEM
ML1024	ML-KEM-1024	L5	KEM
HQ128	HQC-128	L1	KEM
HQ192	HQC-192	L3	KEM
HQ256	HQC-256	L5	KEM
MC348	Classic-McEliece-348864	L1	KEM
MC460	Classic-McEliece-460896	L3	KEM
MC8192	Classic-McEliece-8192128	L5	KEM
DSA44	ML-DSA-44	L1	SIG
DSA65	ML-DSA-65	L3	SIG
DSA87	ML-DSA-87	L5	SIG
F512	Falcon-512	L1	SIG
F1024	Falcon-1024	L5	SIG
SP128	SPHINCS ⁺ -SHA2-128s	L1	SIG
SP192	SPHINCS ⁺ -SHA2-192s	L3	SIG
SP256	SPHINCS ⁺ -SHA2-256s	L5	SIG
AESG	AES-256-GCM	—	AEAD
CH20	ChaCha20-Poly1305	—	AEAD
ASC	Ascon-128a	—	AEAD

I. RESULTS AND ANALYSIS

All measurements were collected on a Raspberry Pi 4 Model B Rev 1.5 (BCM2711, four Cortex-A72 cores at 1.8 GHz, 4 GB RAM) running Debian 12 with kernel 6.12.47 and the performance CPU governor. Power was sampled via an INA219 current sensor (0.1 Ω shunt) at up to 1 kHz with a documented V_{BUS} gain correction of 1.22. Timing used `time.perf_counter_ns()` with garbage collection disabled. Board-level idle power measured $P_{idle} \approx 3.32$ W (30s capture); reported power values include this baseline unless stated otherwise. The Cortex-A72 does *not* implement ARMv8 Crypto Extensions; AES-GCM therefore executes in software table-lookup mode, while ChaCha20-Poly1305 exploits 128-bit NEON SIMD.

Table I defines the short aliases used throughout. Implementations: liboqs-python 0.14.0 (KEM/SIG), OpenSSL 3.5.4 (AES-256-GCM, ChaCha20-Poly1305), ascon-c v1.2 opt64 via ctypes (Ascon-128a, KAT-verified).

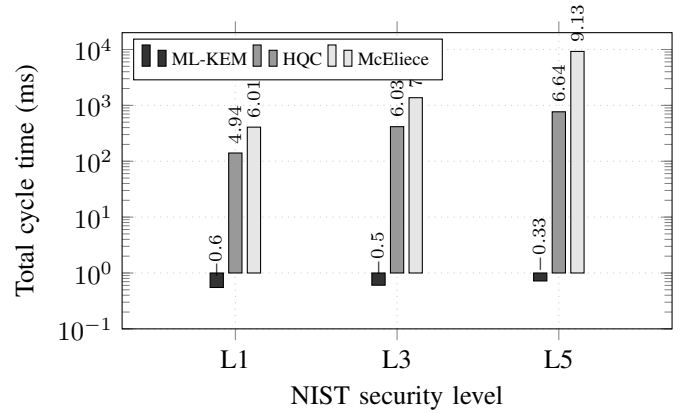


Fig. 1. Total KEM cycle time (keygen+encaps+decaps) by NIST level, log scale. ML-KEM completes in < 1 ms at all levels; HQC and McEliece require 255–12 823 $\times$ longer. 100 iterations, RPi4 @ 1.8 GHz.

A. KEM Primitive Evaluation

Fig. 1 compares total KEM cycle time (keygen+encaps+decaps) across all nine primitives (100 iterations, 1 kHz INA219 sampling).

ML-KEM completes a full cycle in under 0.72 ms at Level 5 (0.550 ms at L1, 0.604 ms at L3), drawing 3.46–3.48 W. HQC’s code-based design incurs 255–1 066 $\times$ higher cost, driven by decapsulation (73–393 ms). Classic McEliece exhibits extreme keygen asymmetry: MC8192 requires 9.02 s for keygen but only 1.89 ms for encapsulation, making it practical only with pre-provisioned keys. Only ML-KEM enables sub-second rekey on ARM Cortex-A72.

B. Signature Primitive Evaluation

Table II evaluates eight signature schemes. Keygen is a one-time cost (0.4–282 ms for ML-DSA/Falcon, 187–282 ms for SPHINCS⁺); the per-handshake sign and verify determine rekey latency.

Falcon achieves the lowest combined sign+verify latency at L1 (0.970 ms) and L5 (1.732 ms), outperforming ML-DSA by 36% and 33% respectively, despite 51–73 $\times$ slower keygen (amortized over key lifetime). SPHINCS⁺ imposes prohibitive signing latency: SP128 at 1.47 s (6.01 J per signature) and

TABLE II
SIGNATURE SIGN+VERIFY ON RP14 (100 ITER, 1 KHz INA219)

Alias	Lvl	Sign (ms)	Verify (ms)	E_s (μ J)	E_v (μ J)	ρ
DSA44	L1	1.179	0.331	4 264	1 175	$1.00\times$
F512	L1	0.778	0.192	2 773	683	$0.64\times$
SP128	L1	1 468	1.582	6 008 719	5 690	$972\times$
DSA65	L3	1.633	0.484	5 849	1 723	$1.00\times$
SP192	L3	2 610	2.299	10 628 558	8 318	$1 234\times$
DSA87	L5	1.847	0.723	6 612	2 564	$1.00\times$
F1024	L5	1.452	0.280	5 175	995	$0.67\times$
SP256	L5	2 321	3.292	9 516 164	11 738	$904\times$

ρ : sign+verify cost vs. ML-DSA at same level. Falcon $\rho < 1$ = faster combined sign+verify. SP128 sign = $1.47\text{ s} \cdot 6.01\text{ J}$ per invocation.

TABLE III
AEAD PERFORMANCE AT 256-BYTE PAYLOAD (10 000 ITERATIONS, RP14)

Alias	Enc (μ s)	Dec (μ s)	P_{enc} (W)	P_{dec} (W)	$E_{256\text{B}}$ (μ J)	E_{bit} (nJ/bit)
AESG	9.25	9.09	3.63	3.57	66.0	32.2
CH20	6.47	6.05	3.54	3.58	44.6	21.8
ASC	12.72	12.85	3.84	3.59	95.0	46.4

$E_{256\text{B}} = P_{\text{avg}} \times (t_{\text{enc}} + t_{\text{dec}})$. $E_{\text{bit}} = E_{256\text{B}}/2048$. Board-level power; $P_{\text{idle}} = 3.320\text{ W}$ not subtracted. Crypto-only ΔP : AESG 0.31, CH20 0.22, ASC 0.52 W above idle.

SP192 at 2.61 s (10.63 J), approaching the rekey period at short intervals. Verification remains fast ($\leq 3.3\text{ ms}$) for all algorithms, confirming the GCS verification path is never the bottleneck. Falcon is therefore the preferred signature scheme where its non-standard keygen model is acceptable; ML-DSA provides the general-purpose default at under 2.6 ms combined cost through Level 5.

C. AEAD Primitive Evaluation

AEAD selection operates exclusively on the UDP data plane and does *not* affect handshake timing. Table III reports results from 10 000 iterations per configuration (500-iteration warmup, two-pass methodology: timing pass with GC disabled, power pass with INA219 at $\sim 103\text{ Hz}$).

Timing distributions are concentrated: $\sigma_{\text{enc}} = 0.38\text{ }\mu\text{s}$ (AESG) and $0.77\text{ }\mu\text{s}$ (ASC). CH20 exhibits $\sigma = 3.99\text{ }\mu\text{s}$ due to rare OS scheduling jitter (P95=6.46, P99=6.69 μs over 10 000 samples; max=208 μs), with median 6.30 μs confirming the mean is representative.

Table IV and Fig. 2 show scaling across payload sizes spanning MAVLink telemetry (9–263 B) to bulk transfer (4096 B).

At 4096 B the CH20:AESG throughput ratio reaches $3.93\times$, while below 64 B setup cost dominates and the ciphers converge. The energy-per-bit metric at 256 B yields CH20=21.8 nJ/bit, AESG=32.2 nJ/bit, ASC=46.4 nJ/bit. For a typical 50 Hz MAVLink stream (263 B packets), per-second AEAD cost ranges from 2.23 mJ (CH20) to 4.75 mJ (ASC)—negligible versus the 3.32 W board draw. ChaCha20-Poly1305 is therefore the optimal data-plane cipher for ARM cores

TABLE IV
AEAD ENCRYPT LATENCY SCALING BY PAYLOAD SIZE (μ s)

Alias	9 B	33 B	64 B	256 B	1024 B	4096 B
AESG	4.75	5.45	5.96	9.25	22.98	75.71
CH20	5.30	5.38	5.33	6.47	9.38	19.25
ASC	11.38	11.82	11.78	12.72	17.27	30.86

Per-KiB slopes: CH20 3.4, ASC 4.8, AESG 17.7 $\mu\text{s}/\text{KiB}$.

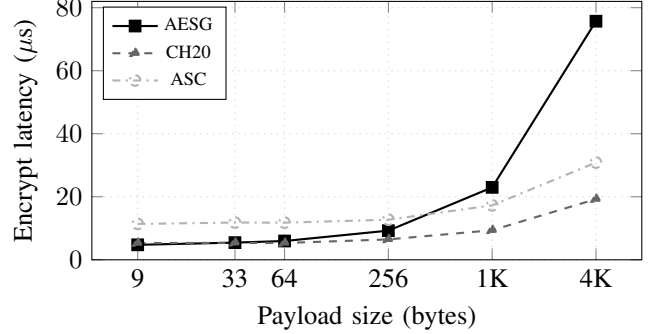


Fig. 2. AEAD encrypt latency versus payload size. CH20 (NEON SIMD) scales at $3.4\text{ }\mu\text{s}/\text{KiB}$; AESG at $17.7\text{ }\mu\text{s}/\text{KiB}$ (software table-lookup). Below 64 B, setup cost dominates and ciphers converge.

without hardware AES acceleration, delivering 32% lower energy per bit than AES-256-GCM.

D. Suite-Level Evaluation

The crypto-only handshake cost is determined by the KEM and SIG selection:

$$T_{\text{HS}} = t_{\text{keygen}} + t_{\text{encaps}} + t_{\text{decaps}} + t_{\text{sign}} + t_{\text{verify}} \quad (1)$$

Table V presents boundary suites per NIST level (fastest, worst-KEM, worst-SIG), with T_{HS} reconstructed from primitive benchmarks to isolate cryptographic cost. Fig. 3 visualizes the four-order-of-magnitude cost range.

The default suite (ML768+DSA65) achieves 2.72 ms crypto-only handshake at Level 3. At Level 5, ML1024+F1024 requires only 2.45 ms—faster than the Level 3 default because Falcon-1024 verify (0.280 ms) is 42% faster than DSA65 verify (0.484 ms). For ML-KEM suites the signature accounts for 73–99% of T_{HS} ; for HQC and McEliece suites, the KEM dominates at 98–99%.

System-level note. The crypto-only $T_{\text{HS}} = 2.72\text{ ms}$ isolates pure cryptographic cost. End-to-end handshake measurements (Section I-E) report $T_{\text{base}} = 13.01\text{ ms}$ for the same suite; the 10.3 ms gap is attributable to protocol framing, HKDF key derivation, and Python interpreter overhead. ML-KEM+Falcon achieves sub-2.5 ms at Level 5, establishing lattice-based PQC as the only viable family for real-time UAV rekey with ρ gaps of $314\text{--}3\,770\times$ to alternatives.

E. DDoS–Cryptographic Interaction

Concurrent DDoS detection competes with the cryptographic pipeline for CPU and thermal headroom. All 72

TABLE V
SUITE HANDSHAKE COST BOUNDARIES (FROM PRIMITIVES)

KEM	SIG	T_{HS} (ms)	E_{HS} (mJ)	ρ
<i>L1</i>				
ML512	F512	1.52	5.3	1.0×
MC348	F512	408.4	1 579	269×
ML512	SP128	1 470	5 737	967×
<i>L3 — Default (*)</i>				
ML768	DSA65	2.72	9.6	1.0×
MC460	DSA65	1 374	5 358	505×
ML768	SP192	2 613	10 290	961×
<i>L5</i>				
ML1024	F1024	2.45	8.6	1.0×
MC8192	DSA87	9 235	36 800	3 770×
ML1024	SP256	2 325	9 105	949×

$T_{\text{HS}} = t_{\text{KEM}} + t_{\text{SIG}}$ (crypto-only, no network). $E_{\text{HS}} \approx P_{\text{avg}} \times T_{\text{HS}}$. ρ : relative to fastest at same level. (*) Default: cs-mlkem768-aesgcm-mldsa65.

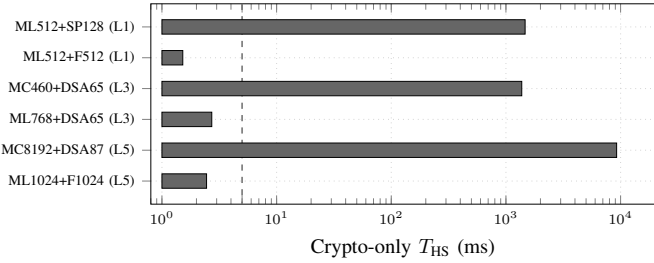


Fig. 3. Suite handshake crypto cost (log scale). ML-KEM suites cluster below 5 ms (dashed line) across all NIST levels; McEliece and SPHINCS⁺ alternatives span three to four orders of magnitude higher. ML768+DSA65 is the system default.

level-matched suites were benchmarked under baseline (no detector), XGBoost, and TST scenarios (10 s per suite per scenario, crypto-only handshakes).¹

XGBoost adds a median 0.05% overhead (mean 2.5%) to PQC handshakes, confirming tree-based detection is compatible with real-time cryptographic operations. TST nearly triples the default suite latency (13.01 $\rightarrow$ 37.90 ms, +191%), driven by its 93% CPU utilization starving the handshake of execution time.

Table VII and Fig. 4 compare the four trained CIC-IoT-2023 detectors on inference cost.

XGBoost achieves the highest reported accuracy (94.55%) at 7.42 ms mean latency and 31.7 mJ per inference. Random-Forest consumes 10 $\times$ more energy (291.4 mJ) due to its 200-tree ensemble. TST delivers the lowest accuracy (90.27%) at the highest power cost (+1 973 mW, 93% CPU), with a P99 tail of 26.12 ms indicating variable inference latency. Tree-based models are therefore the preferred architecture for embedded PQC-DDoS co-deployment: XGBoost adds negligible handshake overhead while providing the highest detection accuracy.

¹Both lightweight detectors use 2 features (Mavlink_Count, Total_length), distinct from the 54-feature CIC-IoT-2023 models in Table VII. TST uses seq_len=1; functionally a 3-layer MLP with 1.6 M parameters.

TABLE VI
PQC HANDSHAKE OVERHEAD UNDER CONCURRENT DETECTION (72 SUITES)

Statistic	XGBoost	TST
Mean ΔT_{HS} (%)	+2.5	+71.8
Median ΔT_{HS} (%)	+0.05	+32.6
Max ΔT_{HS} (%)	+137.2	+405.1
<i>Default suite: ML768+AESG+DSA65</i>		
System-level T_{base} (ms)	13.01	
T_{HS} + detector (ms)	13.46	37.90
ΔT_{HS} (%)	+3.5	+191.3

$\Delta T_{\text{HS}} = (T_{\text{det}} - T_{\text{base}})/T_{\text{base}} \times 100$. System-level T_{base} includes protocol overhead beyond crypto-only T_{HS} (cf. Table V).

TABLE VII
DDoS DETECTOR INFERENCE ON RPi4 (CIC-IoT-2023)

Model	Acc. (%)	F1	t_{inf} (ms)	P99 (ms)	ΔP (mW)	E_{inf} (mJ)
LightGBM	93.47 [†]	0.937 [†]	6.72	8.59	+999	29.0
XGBoost	94.55 [†]	0.947 [†]	7.42	9.52	+951	31.7
RF	93.35 [†]	0.935 [†]	74.68	102.63	+594	291.4
TST	90.27 [†]	0.80 [†]	10.72	26.12	+1 973	56.7

[†]Accuracy and F1 from model documentation; inference metrics measured on RPi4 (15 s per model, idle baseline 3.309 W). CIC-IoT-2023: 3.4 M samples. XGBoost/LightGBM/RF: 54 features, 15 classes. TST: 46 features, 6 classes.

F. Scheduler Behavioral Impact

The Multi-Dimensional Energy-Aware Scheduler (MDEAS) manages three decision axes evaluated at 1 Hz: (1) AEAD cipher selection based on energy and thermal state; (2) security-level promotion/demotion triggering PQC rekey; and (3) DDoS detector activation based on threat telemetry and thermal headroom.

Each security-level transition requires a complete PQC handshake. The rekey overhead fraction is:

$$\Phi(R) = \frac{T_{\text{HS}}}{R + T_{\text{HS}}} \quad (2)$$

where R is the rekey interval. For the default suite ($T_{\text{HS}} = 2.72$ ms) at the minimum 30 s interval: $\Phi = 9.1 \times 10^{-5}$ (<0.01%). Even the heaviest suite (MC8192+SP256, $T_{\text{HS}} \approx 11.6$ s) at 30 s yields $\Phi = 0.279$ (27.9%), which the break-even calculator correctly flags as non-viable.

Table VIII summarizes the per-detector resource profiles used for proactive thermal management.

The MDEAS enforces cross-axis constraints to prevent pathological configurations: Ascon-128a is forbidden when TST is active, as its higher per-packet cost combined with 93% CPU utilization exceeds the real-time budget; McEliece KEMs are discouraged under TST due to CPU-bound key-gen contention; all McEliece $\times$ SPHINCS⁺ pairs are forbidden because T_{HS} would exceed the 45 s rekey timeout; and detector activation is blocked when temperature exceeds the per-detector T_{max} ceiling. These constraints reduce the 648 theoretical configurations (216 suites $\times$ 3 detector states)

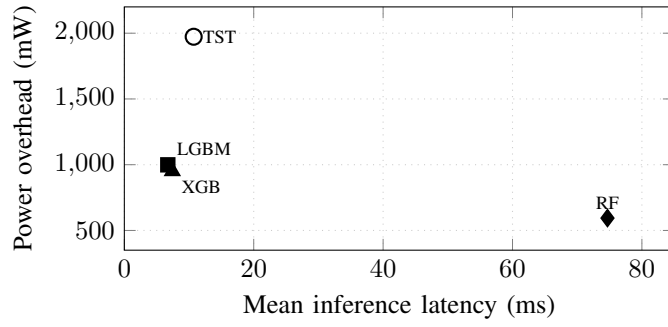


Fig. 4. Detector resource footprint: inference latency vs. power overhead. XGBoost and LightGBM cluster in the low-latency, moderate-power region. RF is a latency outlier (10 $\times$); TST doubles power draw at modest latency.

TABLE VIII
MDEAS DETECTOR OVERHEAD PARAMETERS

Detector	ΔP (W)	ΔT ($^{\circ}\text{C}$)	ΔCPU (pp)	$T_{\max}$ ($^{\circ}\text{C}$)
None	0.00	0.0	0	80.0
XGBoost	0.95	4.8	35	75.0
TST	1.97	10.7	91	69.0

$T_{\max}$: baseline temperature ceiling before detector activation is blocked (80 $^{\circ}\text{C}$ throttle minus ΔT). ΔCPU in percentage points.

to approximately 500 viable operating points, navigating the security–energy–thermal Pareto surface via 1 Hz telemetry feedback while guaranteeing the configured minimum security level.